

Brownian Motion

Monte Carlo Statistical Methods

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1 Brownian Motion as the Scaling Limit of a Simple Random Walk

1.1 Simple Random Walk

Let $\{X_k\}_{k \geq 1}$ be i.i.d. random variables defined by

$$X_k = \begin{cases} +1 & \text{with prob. } \frac{1}{2}, \\ -1 & \text{with prob. } \frac{1}{2}. \end{cases}$$

Define the random walk

$$S_0 = 0, \quad S_n = \sum_{k=1}^n X_k.$$

Basic properties:

$$\mathbb{E}[S_n] = 0, \quad \text{Var}(S_n) = n.$$

Intuition: after n steps, the walker experiences n independent unit kicks, so its typical spread scales like

$$|S_n| \approx \sqrt{n},$$

the first hint of diffusive behavior.

1.2 Diffusive Scaling

To obtain a continuous-time limit, we compress n steps into $[0, 1]$ by setting

$$t = \frac{k}{n},$$

and scale the height by $\sqrt{n}$ to keep fluctuations finite:

$$W_n(t) = \frac{S_{\lfloor nt \rfloor}}{\sqrt{n}}.$$

Intuition:

- Time scaling makes the walk dense in the interval.
- Height scaling keeps variance at time t equal to t (not nt).
- This creates a non-exploding, non-degenerate limit.

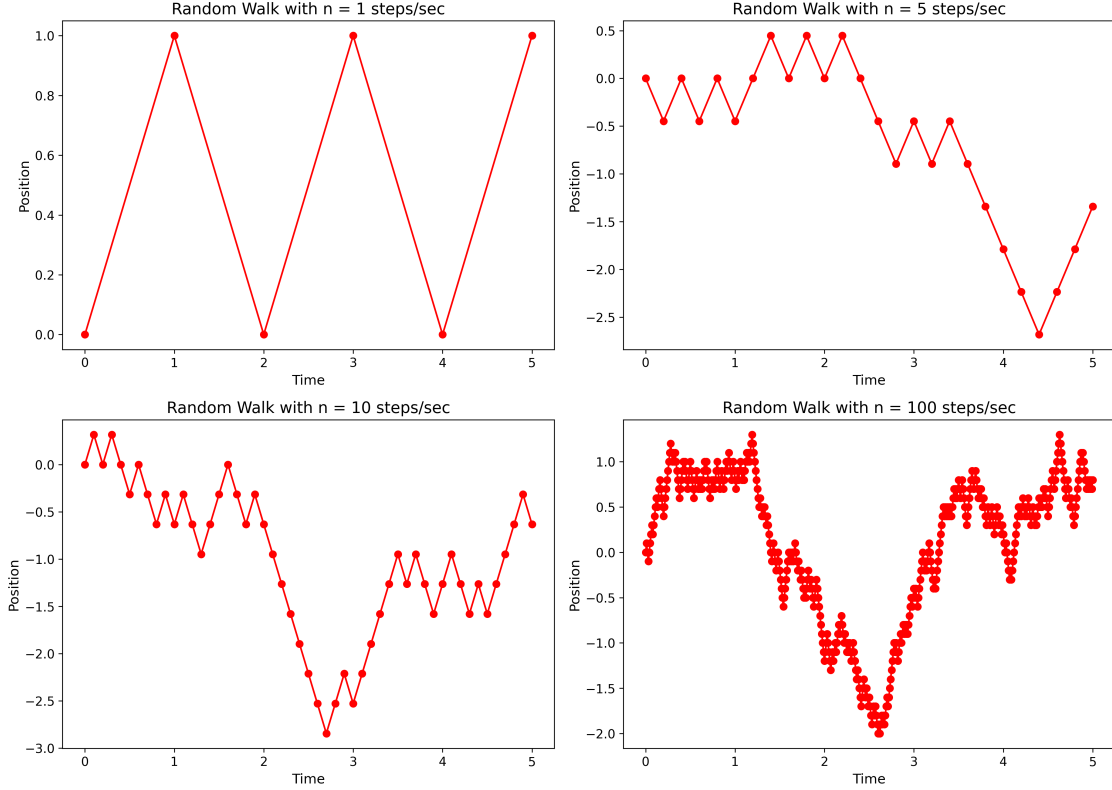


Figure 1: Diffusive scaling of the simple random walk for increasing n . As n grows, jumps shrink and the piecewise-constant path visually approaches a continuous curve. Time is compressed as $t = k/n$ and height is scaled by $1/\sqrt{n}$.

1.3 Increment Structure

Fix $0 \leq s < t \leq 1$. Then

$$W_n(t) - W_n(s) = \frac{1}{\sqrt{n}} \sum_{j=\lfloor ns \rfloor + 1}^{\lfloor nt \rfloor} X_j.$$

Independent increments

Since sums over disjoint index sets involve independent X_j ,

$$W_n(t) - W_n(s) \text{ is independent of } W_n(u) - W_n(v)$$

whenever $(s, t]$ and $(v, u]$ do not overlap.

Mean and variance

$$\mathbb{E}[W_n(t) - W_n(s)] = 0,$$

$$\text{Var}(W_n(t) - W_n(s)) = \frac{\lfloor nt \rfloor - \lfloor ns \rfloor}{n} \longrightarrow t - s.$$

Gaussian limit

By the central limit theorem,

$$W_n(t) - W_n(s) \implies \mathcal{N}(0, t - s).$$

Thus the limiting increments are Gaussian with variance equal to elapsed time.

1.4 Limit Covariance

For $s \leq t$,

$$\mathbb{E}[W_n(s)W_n(t)] = \frac{1}{n}\mathbb{E}[S_{\lfloor ns \rfloor} S_{\lfloor nt \rfloor}] = \frac{\lfloor ns \rfloor}{n} \longrightarrow s.$$

Hence the limiting covariance function is

$$\mathbb{E}[B(s)B(t)] = \min(s, t).$$

This identifies $B(t)$ as standard Brownian motion.

1.5 Continuous Brownian Motion via Euler Simulation

To visualize the limit, we simulate Brownian motion directly using

$$W(t + \Delta t) - W(t) \sim \mathcal{N}(0, \Delta t) = \sqrt{\Delta t} Z_k, \quad Z_k \sim \mathcal{N}(0, 1).$$

The sample path is built via cumulative summation.

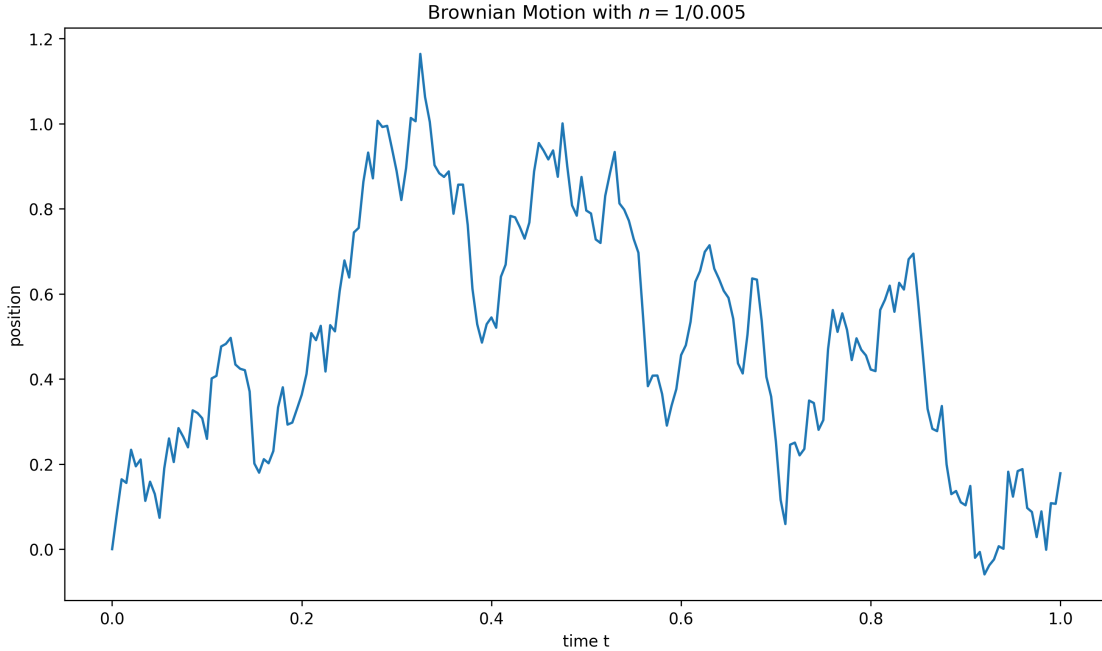


Figure 2: Simulation of a single Brownian path using the Euler scheme. Smaller Δt produces rougher, more accurate paths.

1.6 Variance Growth: $\text{Var}(B(t)) = t$

Since Brownian increments satisfy

$$B(t) - B(s) \sim \mathcal{N}(0, t - s),$$

we have

$$\text{Var}(B(t)) = t.$$

The variance cone $\pm\sqrt{t}$ captures two standard deviations.

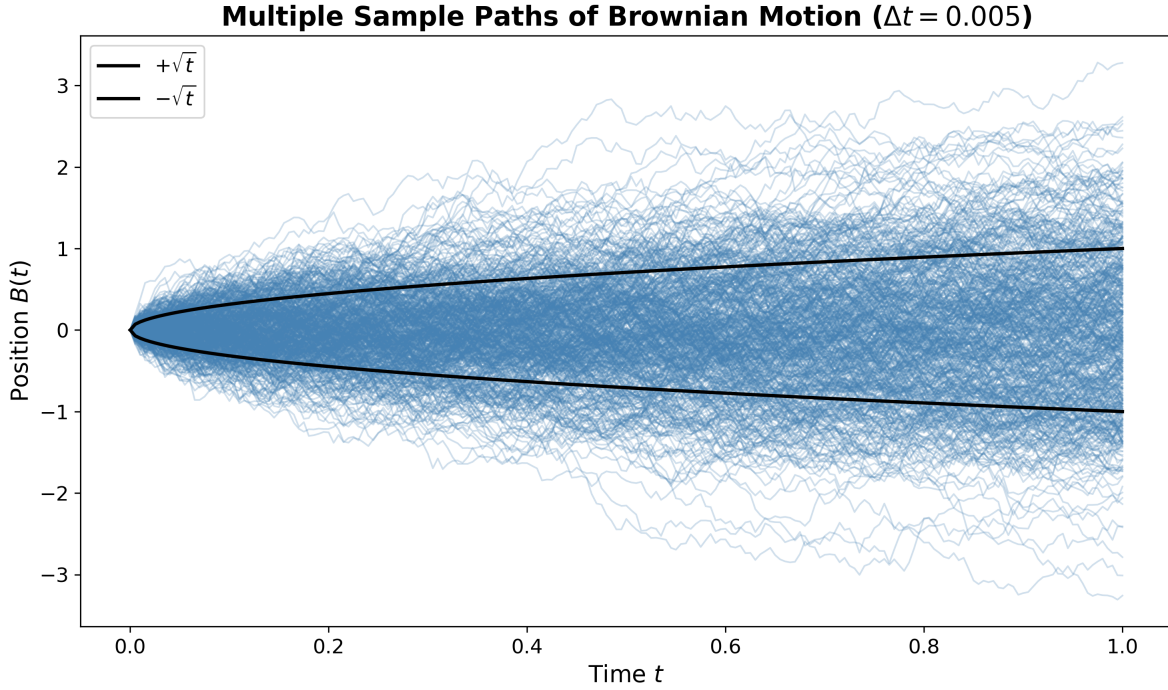


Figure 3: Multiple sample paths of standard Brownian motion $B(t)$ simulated with time step $\Delta t = 0.005$. The black curves show the theoretical $\pm\sqrt{t}$ envelopes corresponding to one standard deviation. Most trajectories remain within this band, illustrating the variance law $\text{Var}(B(t)) = t$.

1.7 Gaussian Marginals

For each fixed t ,

$$B(t) \sim \mathcal{N}(0, t).$$

We verify this by histogramming the simulated values at two times $t_1 < t_2$.

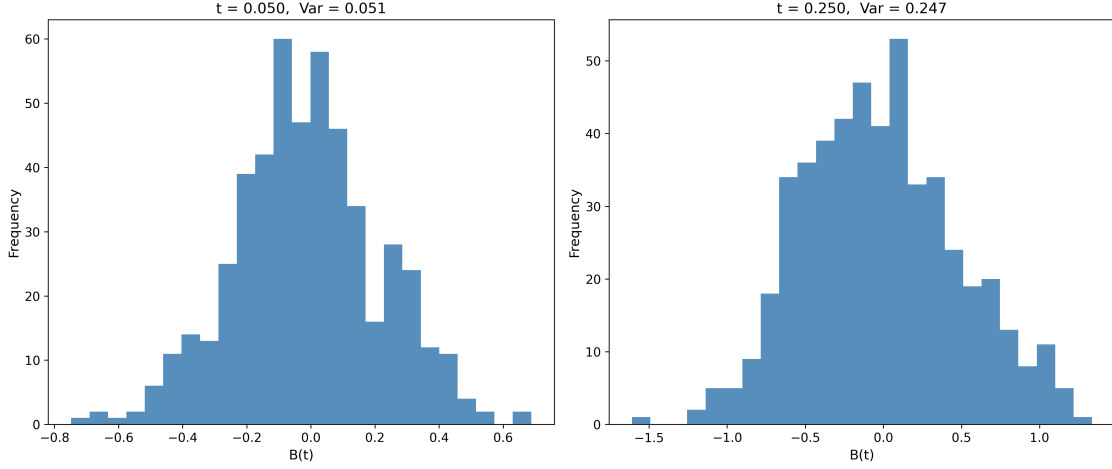


Figure 4: Histograms of $B(t_1)$ and $B(t_2)$ from simulation. Both empirical densities match the corresponding normal distributions $\mathcal{N}(0, t_1)$ and $\mathcal{N}(0, t_2)$.

1.8 Stationary Increments

For Brownian motion,

$$B(t + \Delta) - B(t) \sim \mathcal{N}(0, \Delta),$$

independent of t .

Simulation verifies that all increments of length Δ have the same empirical distribution.

1.9 Self-Similarity

Brownian motion satisfies

$$B(ct) \stackrel{d}{=} \sqrt{c} B(t).$$

This can be confirmed in simulation by comparing $B(2t)/\sqrt{2}$ with $B(t)$.

1.10 Continuity and Roughness

$B(t)$ is continuous everywhere but almost surely nowhere differentiable. As $\Delta t \rightarrow 0$, increments remain of size

$$\sqrt{\Delta t},$$

too large relative to Δt for a derivative to exist.

1.11 Summary of Convergence

The scaled random walk $W_n(t)$ converges in distribution to Brownian motion $B(t)$ because:

- increments become independent,
- increments converge to Gaussian,
- variance and covariance converge to $t - s$ and $\min(s, t)$,
- the sequence is tight, implying continuous limiting paths.

Therefore

$$W_n \Longrightarrow B.$$

1.12 Relevance to Robotics and Bayesian Filtering

Brownian motion models the accumulation of process noise over time. In continuous dynamics:

$$dx = f(x, u) dt + \Sigma dB_t,$$

and discrete-time filters (KF, EKF, UKF, PF) use the approximation

$$w_k \sim \mathcal{N}(0, Q\Delta t),$$

coming directly from Brownian increments.

Thus Brownian motion provides the mathematically correct foundation for uncertainty growth in dynamical systems.